Small-scale fusion tackles energy, space applications

Efforts are underway to exploit a strategy that could generate fusion with relative ease.

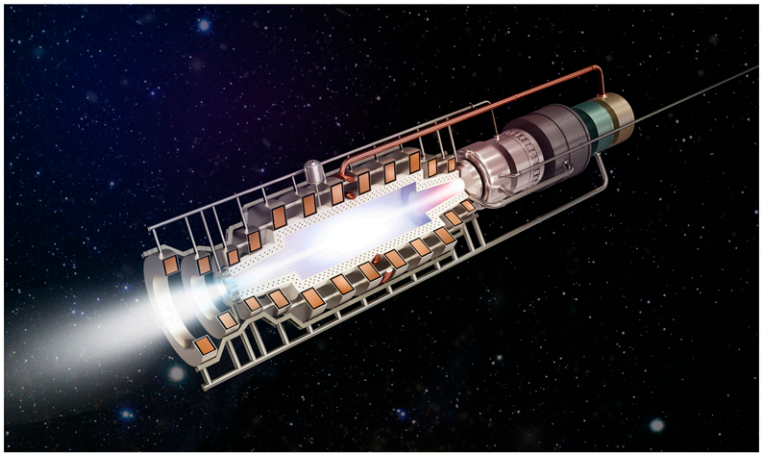
M. Mitchell Waldrop, *Science Writer*

On July 14, 2015, nine years and five billion kilometers after liftoff, NASA's New Horizons spacecraft passed the dwarf planet Pluto and its outsized moon Charon at almost 14 kilometers per second—roughly 20 times faster than a rifle bullet.

The images and data that New Horizons painstakingly radioed back to Earth in the weeks that followed revealed a pair of worlds that were far more varied and geologically active than anyone had thought possible. The revelations were breathtaking—and yet tinged with melancholy, because New Horizons was almost certain to be both the first and the last spacecraft to visit this fascinating world in our lifetimes.

Unless, that is, Samuel Cohen succeeds with the offbeat fusion reactor that he's developing at the Princeton Plasma Physics Laboratory in New Jersey.

Cohen's current prototype is a clear plastic cylinder that sits in the middle of his lab amidst a dense mass of cables, magnets, and power supplies, emitting a violet pulse of light every two seconds like a two-meter-long strobe light. "We're only using hydrogen right now," Cohen explains, referring to the ionized plasma inside the tube that's emitting the flashes. So there are no actual fusion reactions taking place; that's not in his research plan until the mid-2020s, when he hopes to be working with a more advanced prototype at least three times larger than this one.



Samuel Cohen and his team hope to beat the standard timetable for fusion by about a decade using a reactor—initially for rocket propulsion—that's a fraction of the size and cost of the huge tokamak devices. Cohen's design takes advantage of the phenomenon of field reversed configuration (FRC), in which a dense mass of ionized plasma holds itself together. Image credit: Princeton Plasma Physics Laboratory.

If that hope pans out and his future machine does indeed produce more greenhouse gas–free fusion energy than it consumes, Cohen and his team will have beaten the standard timetable for fusion by about a decade—using a reactor that's just a tiny fraction of the size and cost of the huge, donut-shaped "tokamak" devices that have long devoured most of the research funding in this field. The flagship of this tokamak approach, the International Thermonuclear Experimental Reactor (ITER) now under construction in France, will be twice as large as any fusion reactor before it, will cost at least $20 billion to build, and isn't expected to start producing fusion energy until the mid-2030s.

If and when Cohen does reach his fusion energy milestone, he will likely have company. His device is just one of a family of small, alternative reactor projects designed to exploit a phenomenon known as the field-reversed configuration (FRC): a dense mass of ionized plasma that holds itself together something like a smoke ring and that could allow researchers to achieve fusion conditions with comparatively little effort. Among the members of this family are some of the best-known fusion upstarts: firms such as TAE Technologies (formerly TriAlpha Energy) in Foothill Ranch, California, and Helion Energy in Redmond, Washington.

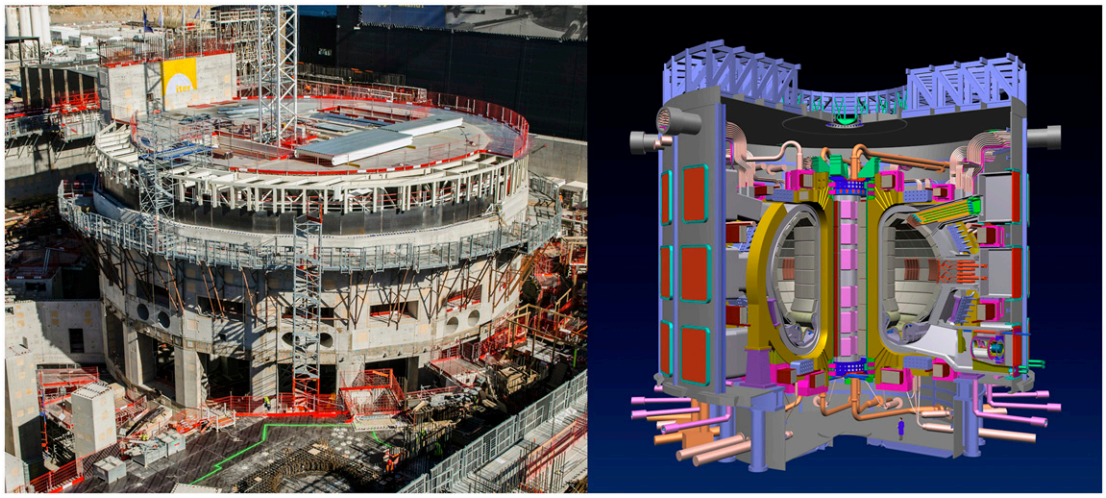
"There's been a rejuvenation in that whole area" of FRCs, says Stephen Dean, a nuclear engineer who has championed fusion energy for more than 50 years. "All of the projects have good ideas, all of them are doing good work." But even if some or all of them do end up producing fusion energy in the lab at some point in the 2020s, he says, all of them are eventually going to have to build a real, power-producing test reactor—something that's not likely to happen for a decade or more.

Pluto Power

That's why Cohen takes the long view. His goal is an ultra-compact reactor that will use a fuel mix containing

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The International Thermonuclear Experimental Reactor (ITER), now under construction in France, will be twice as large as any fusion reactor before it and will cost at least $20 billion to build. To keep the fusion plasma under control, the tokamak design uses strong magnetic fields to guide ionized isotopes around a donut-shaped vacuum chamber. (*Left*) Image credit: Wikimedia Commons/Oak Ridge National Laboratory. (*Right*) Image credit: Science Source/ITER.

helium-3, an isotope that yields a particularly clean form of fusion with minimal radiation risk. But the stuff is exceedingly rare, he says: "So we're not trying to make power for everybody." Instead, the goal is niche uses such as spacecraft propulsion, in which the reactor would fire a very tenuous plasma from one end so that it functions as a rocket (1).

Such a direct fusion drive (DFD) would produce only the most infinitesimal hint of acceleration, says Cohen—about like pushing an 18-wheel truck with your fingertip. But in space, that push would have nothing to resist it. After a year or two, such a rocket could get a 10-ton spacecraft halfway to Pluto, traveling well over 50 kilometers per second.

"Then you'd turn around and decelerate," says Cohen. "And when you got to Pluto, you'd go into orbit." At that point, the reactor would turn off the ion rocket and convert itself into a one-megawatt electrical power source. "Some of that power you can use to send high-definition video back," says Cohen. "And some of it you can beam down to a lander that you've placed on the surface, so it could drive around and drill holes in the ice."

The same type of DFD rockets could also be used to explore the moons of Jupiter and Saturn, says Cohen, or the icy bodies of the Kuiper Belt beyond Pluto, or anywhere else in the outer solar system.

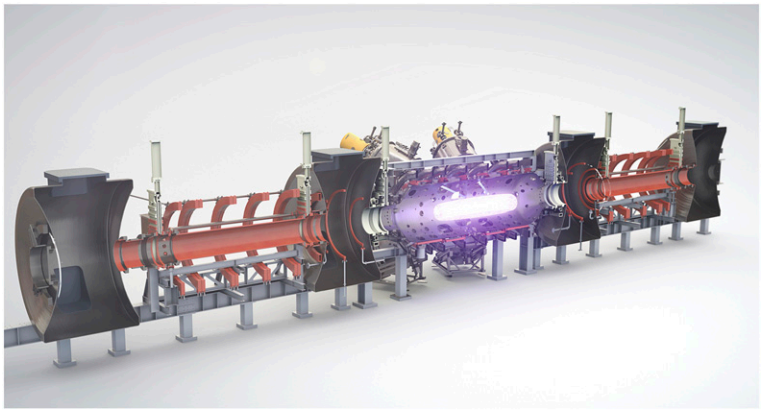
Plasma Problem

Of course, there's a reality check, says Dean: "If you want to make a fusion exhaust system, you still have to be able to make the fusion plasma." It's a trick that neither Cohen nor anyone else has yet managed. Researchers have been trying to harness fusion power since the 1920s and 1930s, when they first realized that stars like the sun get their energy from thermonuclear reactions at their core. And yet, as the many delays and cost overruns on ITER have made clear, success is still years away at best.

Still, old hands like Cohen know the pitfalls of fusion research as well as anyone. Until the late 1990s, his professional life revolved around ITER, which is supposed to be the ultimate expression of the oldest and most promising approach to fusion energy: magnetic confinement. In theory, this is just a matter of ionizing an appropriate mix of light isotopes, trapping them in a magnetic field, and heating them to millions of degrees while simultaneously squeezing them to densities approximating the sun's core. The isotopes will then start fusing into larger nuclei while releasing vast amounts of energy.

In practice, though, hot, ionized plasma doesn't like being confined by a magnetic field; it twists and tries to escape like a living thing. Thus the appeal of the tokamak design, which was a major breakthrough when Soviet physicists introduced it in the 1960s. Thanks to strong magnetic fields that guide the ionized isotopes around and around its donut-shaped vacuum chamber, a tokamak could keep the plasma under control better than almost anything else at the time. And thus the funding agencies' willingness to keep sinking billions of dollars into ITER: a gargantuan tokamak whose 23,000-ton weight will be three times that of the Eiffel Tower, and whose 29 by 29-meter vacuum chamber will be as tall as a seven-story building. This is the scale that a tokamak will need to achieve the elusive goal of "break-even," in which the plasma produces more fusion energy than the machine requires to operate.

Except that to Cohen and an increasing number of other fusion researchers, ITER has laid bare the tokamak's many drawbacks as a practical power source. These start with the facility's size, cost, and complexity, which are so far beyond what power companies are willing to accept that they have all but given up on



TAE Technologies is designing a small fusion reactor capped on each end with electromagnetic cannons pointed barrel to barrel. To start the reaction, each cannon fires a ring of plasma into a central chamber, where the rings merge into a single, furiously spinning FRC. A beam of neutral atoms coming in from the side will simultaneously heat the FRC, supply it with fresh fuel, and stabilize it by maintaining the spin rate. Image credit: TAE Technologies.

fusion, says Dean: "I can't even talk to anyone in the utilities who knows what a tokamak is anymore."

And then there's the neutron problem. The physics of tokamaks limits them to burning a mix of the hydrogen isotopes deuterium and tritium. This fuel is by far the easiest to ignite, requiring comparatively low plasma temperatures of about a hundred million degrees Kelvin. But when the two nuclei fuse to form a helium-4 nucleus (two protons plus two neutrons), they eject the leftover neutron at high energy. And because that particle is electrically neutral and can't be controlled with magnetic fields, it ends up smashing into the tokamak's inner walls and wreaking havoc with their structural integrity. So the walls will have to be replaced perhaps once per year—a maintenance burden that no power company wants to shoulder.

A Different Configuration

By the late 1990s such hurdles were spurring Cohen and others to take a fresh look at FRCs, which had been discovered in the 1960s.

The key advantage was that an FRC doesn't keep its plasma in line by brute force, the way a tokamak does. Instead, the FRC plasma is self-organizing. That is, the magnetic fields that hold it together are mostly generated by currents flowing through the plasma itself, rather than in external coils. This self-organizing property can be found in other plasma structures, which have names such as "spheromak" and "dense plasma focus," says Cohen. But all else being equal, FRC plasmas are much hotter and denser than the others.

Once it's set up, an FRC actually looks less like a smoke ring than an elongated American football, or maybe a short cigar. The "field-reversed" name comes from the way magnetic fields curve around the football's outside and then loop backward through its long axis. This structure tends to dissipate in less than a millisecond, unfortunately—one big reason why only a handful of researchers stuck with the FRCs after tokamaks came along. But the appeal remained: Find a way to stabilize the FRC, and the reactor wouldn't have to be much more than a cylindrical vacuum chamber with a comparatively mild magnetic field running down the midline to hold the plasma football in place.

Self-organization also should make it comparatively easy for the dense, hot plasma inside the FRC to reach the threshold required for fusion. And not just deuterium-tritium fusion, either: FRCs could potentially reach the much higher temperatures required to burn aneutronic fuels such as deuterium-helium–3 or proton-boron–11. These reactions emit most of their fusion energy in the form of charged particles such as protons or helium-4 nuclei, which—unlike neutrons—can be captured and controlled with magnetic fields. This would make it much easier to extract energy from the fusion products before they can damage the reactor walls, and would allow the reactor to get by with minimal shielding.

So in principle, says Cohen, FRC-based reactors could solve the tokamak's size, complexity, and neutron problems at a stroke. But to make that work in practice, he says, researchers have had to make a series of critical design choices: how to form, stabilize, and control the FRC, how to heat it, what kind of fusion fuel to use, and so on. "You multiply all those options," he says, "you get roughly 80 different potential FRCs."

TAE has been working on one such option since 1998, when it was founded with the goal of fusing protons with boron-11 nuclei. This pB11 reaction is in some ways the ultimate in neutron-free fusion: Its output is just a triplet of positively charged helium-4 nuclei, which are commonly known as alpha particles (thus the company's original name, TriAlpha.) But the reaction also has some significant downsides. For example, its multibillion-degree threshold for fusion is about 20 or 30 times higher than the temperatures required for the deuterium-tritium reaction that ITER will use. Also, it has about half as much energy yield per fusion event.

So to make pB11 work, TriAlpha's design has to be correspondingly ambitious (2). The idea is to cap the reactor on each end with two electromagnetic cannons pointed barrel to barrel. To start things off, each cannon fires a ring of plasma into a central chamber, where the rings merge into a single, furiously spinning FRC. From there, a beam of neutral atoms coming in from the side will simultaneously heat the FRC, supply it with fresh pB11 fuel, and stabilize it by keeping the spin rate up.

It took TAE until 2012 to demonstrate this whole process in a prototype machine (albeit with a nonfusing hydrogen plasma), says the company's CEO, Michl Binderbauer. "We showed these beautiful experiments where, if you start with the standard FRC and you don't do anything, it dies," he says. "But if you start injecting particles, you slow down the decay and expand how long it lives." Since then, says Binderbauer, the company has shown that this process can sustain the FRC indefinitely—or at least, for the five or 10 milliseconds it

takes the 25-megawatt beam to exhaust the energy that researchers are able to store for each shot.

In a working reactor, of course, that beam power would come from the fusion reaction itself, so that the beam and the FRC could keep going as long as the researchers want. That's a milestone TAE hopes to meet with a pB11-burning prototype well before the end of the 2020s, says Binderbauer. This machine will be roughly the size of four double-decker busses parked end to end, he adds—not small, but still just a fraction of the size of ITER.

In Bellevue, Washington, meanwhile, another FRC-based reactor is under development at Helion Energy, which was founded by University of Washington researchers in 2013. Company officials are not discussing their plans publicly at the moment, but they have been relatively open about their approach via their website and publications.

Helion's reactor, like TAE's, will be a linear tube that uses twin plasma guns to form a stationary FRC in the middle. But instead of trying to sustain the FRC, the Helion device will crush it with an ultrastrong magnetic field until the plasma becomes dense enough and hot enough to fuse. The resulting burst of thermonuclear energy will then cause the ball of plasma to explode outward again, pushing back against the magnetic field and allowing the system to harvest that energy. This cycle will then repeat once per second, generating a steady average power output in much the same way that gasoline explosions do in an internal combustion engine.

The Helion reactor will also differ from TAE's in its choice of fuel. Instead of using pB11, it will burn deuterium and helium-3—an isotope often called a "helion." This reaction requires a temperature of several hundred million degrees, intermediate between deuterium-tritium and pB11. But it, too, is aneutronic: the final products are two charged particles, an alpha and a proton.

Or rather, this fuel is almost aneutronic: It's impossible to keep the deuterium nuclei in the fuel from reacting with each other and producing at least some neutrons. But those neutrons are low energy and comparatively easy to shield against. And for Helion, the deuterium-deuterium side-reactions are a plus: the products are an almost equal mix of a neutron plus helium-3, and a proton plus tritium—a radioactive isotope that will decay into helium-3 with a half-life of 12.3 years. So in principle, Helion's reactor can make its own helium-3 fuel, which is otherwise available only in trace amounts extracted from natural gas fields, or produced as a byproduct in Canadian CANDU fission reactors.

Space Reactor

Cohen, for his part, has been pursuing his Princeton Field Reversed Configuration (PFRC) design since 2002, with a strong emphasis on simplicity and compactness (3). The cylindrical plastic vacuum chamber of his current device, PFRC-2, is only 88 centimeters long, with not a plasma cannon or neutral beam injector in sight. Instead, the FRC is generated via a technique first explored by Austrian and Australian physicists and then refined by Cohen himself. He points to four rectangular copper coils that surround the middle of the tube: one each on its front, back, top, and bottom. Each rectangle, in turn, is divided into two smaller rectangles. The idea, says Cohen, is to drive oscillating currents through these coils in a way that sets up a rotating magnetic field inside the tube: a loop of flux that whirls through the plasma like a flipped coin and drags the plasma particles around and around the waist of the cylinder. In the process, he says, "the fields create, stabilize, and heat the FRC"—all in a single deft maneuver.

Indeed, that's what Cohen routinely demonstrates in his lab: Every two seconds, the hydrogen plasma inside is whipped into an FRC, causing a flash. Each flash lasts for only about eight milliseconds, says Cohen, mainly because longer pulses risk melting the cables that supply the antennas with power. "So for the next machine," he says, "we've got to make better cables"—which should keep the FRCs going indefinitely.

In the meantime, Cohen and his group are working on the main goal of PFRC-2, which is to improve the antennas' ability to heat the plasma. This is crucial, notes Binderbauer: "I know Sam well, we root for each other." But compared with the temperatures that TAE is already working with, he says, "his plasmas are cold." Also, he says, it remains to be seen how well the rotating magnetic field approach will work in a full-scale reactor. "I'm not trying to say that it can't be done," says Binderbauer, "but I think those are some of the things that they're going to have to address."

"If you want to do this really ambitious stuff in the 2030s, you need to be developing new technology now."
—Michael Paluszek

Still, Cohen remains confident. Sometime in the next few years, if things progress as planned, he and his group will replace this machine with PFRC-3: a device twice as large that will hopefully allow them to achieve FRCs lasting as long as 10 seconds, with plasma temperatures on the order of 60 million K. And a few years after that, the plan calls for moving up to PFRC-4, an even larger machine designed to use live fusion fuel at temperatures of 600 million K.

That fuel will be deuterium-helium-3, which Cohen calls "the Goldilocks approach" between deuterium-tritium and pB11. Unlike Helion, however, he plans to keep things simple and forgo any attempt to breed new helium-3. Instead, Cohen will just live with the scarcity of helium-3, and focus on niche applications like spacecraft propulsion—an idea that emerged about a decade ago in discussions with Michael Paluszek, president of a space technology company, Princeton Satellite Systems, New Jersey, which is located just a few miles from Cohen's lab.

Of course, it could be another 10 to 15 years before PFRC-based reactors are working reliably enough

for a multi-year deep-space mission, assuming that they work at all. But then, much the same could be said about any of the alternative fusion designs—or for that matter, ITER. Taking the long view is pretty much a requirement for fusion energy research. "If you want to do this really ambitious stuff in the 2030s," Paluszek says, alluding to DFD missions, "you need to be developing new technology now."

1 S. Thomas, Fusion-enabled Pluto orbiter and lander (NASA, 2016). https://www.nasa.gov/feature/fusion-enabled-pluto-orbiter-and-lander/. Accessed September 24, 2019.

2 H. Gota *et al.*, Formation of hot, stable, long-lived field-reversed configuration plasmas on the C-2W device. *Nucl. Fusion* **59**, 112009 (2019).

3 C. Brunkhorst, B. Berlinger, N. Ferraro, S. A. Cohen, The Princeton FRC Rotating-Magnetic-Field-Experiment RF System in 2007 IEEE 22nd Symposium on Fusion Engineering, (2007), pp. 1–4.